

JEE ADVANCED · PRACTICE SET II

KINEMATICS

Class XI · Physics | Nine Education IIT Academy | Lectures 8, 10–12

15 questions · Suggested time: 50 minutes

Name: _____ Roll No.: _____ Date: _____

Take $g = 10 \text{ m/s}^2$ unless stated otherwise. Marking scheme is given at the end of the paper.

SECTION A · MULTIPLE OPTIONS CORRECT

One or more options may be correct.

Q1. A particle moves in the xy -plane with $x = 3t$ and $y = 4t - 5t^2$ (SI units). Which of the following are correct?

- (A) The initial speed is 5 m/s.
- (B) The acceleration is constant, of magnitude 10 m/s^2 .
- (C) The path is a parabola.
- (D) The particle returns to the x -axis at $t = 1.25 \text{ s}$.

Q2. A particle moves along the x -axis with velocity $v = 4\sqrt{x}$ (SI), and is at $x = 1 \text{ m}$ at $t = 0$. Which are correct?

- (A) The acceleration is constant and equal to 8 m/s^2 .
- (B) The speed at $t = 0$ is 4 m/s.
- (C) The particle is at $x = 9 \text{ m}$ at $t = 1 \text{ s}$.
- (D) The acceleration increases with x .

Q3. A particle with initial velocity 20 m/s decelerates as $a = -2v$ (SI). Which are correct?

- (A) Its velocity is 10 m/s at $t = \frac{1}{2} \ln 2 \text{ s}$.
- (B) The total distance travelled before it stops is 10 m.
- (C) The particle comes to rest in a finite time.
- (D) It covers 5 m in the first $\frac{1}{2} \ln 2$ seconds.

Q4. A projectile is launched from level ground with speed u at angle θ . Over the complete flight:

- (A) The magnitude of the average velocity is $u \cos \theta$.
- (B) The magnitude of the change in velocity is $2u \sin \theta$.
- (C) The radius of curvature of the path at the highest point is $\frac{u^2 \cos^2 \theta}{g}$.
- (D) The average acceleration is zero.

Q5. A stone is thrown horizontally at 15 m/s from a 45 m cliff ($g = 10 \text{ m/s}^2$).

- (A) It lands after 3 s.
- (B) It lands 45 m from the base of the cliff.
- (C) Its impact speed is $15\sqrt{5} \text{ m/s}$.
- (D) Its impact velocity makes 45° with the horizontal.

Q6. Two particles are projected simultaneously from the same point with the same speed u but at different angles θ_1 and θ_2 . Neglect air resistance.

- (A) The velocity of one relative to the other is constant in magnitude and direction.
- (B) The path of one as seen from the other is a straight line.
- (C) Their separation increases linearly with time.
- (D) They will meet again in mid-air.

SECTION B · MATRIX MATCHING

Q7. A projectile is launched from level ground with fixed speed u . Match the projection angle with its property.

Column I	Column II
(A) $\theta = 30^\circ$	(P) Range is maximum
(B) $\theta = 45^\circ$	(Q) Maximum height = $\frac{1}{4}$ of the range
(C) $\theta = 60^\circ$	(R) Range is zero
(D) $\theta = 90^\circ$	(S) Time of flight is maximum
	(T) Another angle in Column I gives the same range

Q8. Match the given kinematic relation ($k > 0$, motion along the x -axis) with its consequence.

Column I	Column II
(A) $v = k\sqrt{x}$	(P) Acceleration is constant
(B) $v = -kx$	(Q) Speed decays exponentially with time
(C) $a = -kv$	(R) Motion is simple harmonic
(D) $a = -kx$	(S) The v - x graph is a straight line
	(T) The v - x graph is not a straight line

Q9. A particle is launched from the top of a 20 m tower with speed 10 m/s ($g = 10 \text{ m/s}^2$). Match the launch direction with the correct impact data.

Column I	Column II
(A) Horizontal	(P) Time of flight = 2 s
(B) Vertically downward	(Q) Impact speed = $10\sqrt{5} \text{ m/s}$
(C) Vertically upward	(R) Time of flight = $(1 + \sqrt{5}) \text{ s}$
(D) 30° above horizontal	(S) Horizontal distance covered is zero
	(T) Time of flight exceeds 2 s

SECTION C · NUMERICAL VALUE TYPE

Give the answer correct to two decimal places.

Q10. A particle starts from rest with $a = (6 - 2t) \text{ m/s}^2$. Find the distance travelled (in m) until it next comes momentarily to rest.

Q11. A particle moves with $a = 4x$ (SI) and has $v = 3 \text{ m/s}$ at $x = 0$. Find its speed (in m/s) at $x = 2 \text{ m}$.

Q12. A projectile is fired at 20 m/s at 60° to the horizontal ($g = 10 \text{ m/s}^2$). Find the magnitude of the change in its velocity (in m/s) between launch and the highest point.

Q13. A ball is projected at 25 m/s at 53° above the horizontal ($\sin 53^\circ = 0.8$, $\cos 53^\circ = 0.6$, $g = 10 \text{ m/s}^2$). Find the time interval (in s) between the two instants at which its velocity makes 45° with the horizontal.

Q14. A ball is projected at 10 m/s at 30° above the horizontal from the top of a 10 m tower ($g = 10 \text{ m/s}^2$). Find the horizontal distance (in m) from the foot of the tower where it strikes the ground.

Q15. The trajectory of a projectile is $y = \sqrt{3}x - x^2/20$ (x, y in metres, $g = 10 \text{ m/s}^2$). Find the speed of projection (in m/s).

Marking. Section A — +4 all correct; +3/ +2/ +1 partial; -2 for any wrong option. Section B — +2 per correct row, -1 per wrong row. Section C — +4 correct, 0 wrong.